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# SOFTWARE REQUIREMENTS SPECIFICATIONS

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# 1 Member Contribution Assessment

| ID | Name | Role(s) | Contribution (%) | Evidence |
|---|---|---|---|---|
| 24127278 | Hoàng Minh Huy | <Team lead / PM> | 18 | https://github.com/Ne1ntendo/InfraLine/issues/20 |
| 24127048 | Nguyễn Thế Huy | <Business analyst> | 23 | https://github.com/Ne1ntendo/InfraLine/issues/22 |
| 24127450 | Nguyễn Chí Minh | <Designer> | 18 | https://github.com/Ne1ntendo/InfraLine/issues/23 |
| 24127329 | Lê Nguyễn Quốc Bảo | <Developer> | 18 | https://github.com/Ne1ntendo/InfraLine/issues/24 |
| 24127142 | Đoàn Minh Tuấn | <Tester> | 23 | https://github.com/Ne1ntendo/InfraLine/issues/21 |

# 2 Introduction

## 2.1 Purpose and Scope

This Software Requirement Specifications (SRS) document is used to describe the functional and non-functional requirements of Infraline, a web-based infrastructure maintenance management system. It informs multiple parties of an organization like employers, employees and technical support team about the current state of the infrastructure and their expected state to request support should anything happen. This document defines the system's expected behaviors, features, constraints and quality requirements which also serves as reference for the development, design and testing phases, ensuring that all team members share a common understanding of the system.

**In Scope:**

- Infrastructure reporting: Requesters can submit a report about broken infrastructure through a forum.
- Communication channels: Support chatting between the requester and the technician.
- Location ping: Requester can narrow down the location of the broken infrastructure for earliest support.
- Maintenance history: Users can see the history of reports and the infrastructures fixed.
- Maintenance status: Users can see the status of the problems reported whether it's fixed or still being worked on.
- Maintenance team checklist: Maintenance team can have a checklist of work orders they are tasked to deal with.
- Email response: A response through email will be sent to the requester when the problem has been dealt with.
- Report tags: Requesters can tag the problem as mechanical, electrical,... so that the maintenance team can have an easier time to judge and prioritize work orders.
- Set priority: Administrators can set the priority to reports posted on the forum to prioritize infrastructures which serve crucial purposes.

- Report moderation: Administrators can accept or reject a report if the problem isn't worth addressing or out of scope.

**Out of scope:**

- Mobile support.
- AI assistant.
- 2D / 3D map of the facility for location ping.

**Intended readers:**

- Project Manager: To understand and monitor the project development process.
- Business Analyst: To analyze, document and maintain system requirements.
- Designer: To reference system architecture, database and UI design.
- Developer: To understand the system's behavior and implement required features.
- Testers: To write test cases and verify whether the system meets its requirements.
- Stakeholders: To understand the system's capabilities and determine whether it meets their needs.

## 2.2 Glossary

| Term | Definition |
|---|---|
| SRS (Software requirements specifications) | A document that describes functional and non-functional requirements of a software. |
| Infrastructure | Physical facilities, equipment and systems used at a work place or education institution. |
| Stakeholder | An individual or organization that has an interest or is affected by the system. |
| Functional Requirement | Requirements specifying the function or behavior that the system must provide. |
| Non-functional Requirement | Requirements specifying the quality attribute or constraints of the system like security, reliability,... |
| Administrator | The user that is responsible for managing the system, moderate reports and overseeing maintenance progress. |
| Use case | Description of interaction between an actor and the system to achieve a specific goal. |
| Actor | A user or an external system that interacts with the application. |
| MoSCoW | A prioritization technique that categorizes requirements into Must-have, Should-have, Could-have, and Won't-have. |
| Traceability Matrix | A table that maps requirements to related use cases, design elements, and test cases. |

# 3 Problem Statement

In workplaces and educational institutions, infrastructure and equipment such as factory machinery, electrical systems and classroom equipment play an important role in daily operations. When these facilities break down or become faulty, the issues need to be reported to the appropriate personnel so that maintenance work can be carried out.

The current workflow of most facilities involves several parties with different responsibilities. Workers and students are responsible for identifying and reporting infrastructure issues. Management staff oversee the reported issues, approve submissions and assign maintenance personnel. Maintenance staff are responsible for inspecting and resolving the reported problems.

To support this workflow, InfraLine is proposed as a web-based work order management system. It provides a common platform for reporting infrastructure issues, managing work orders and monitoring maintenance progress

By bringing these activities into one system, InfraLine aims to establish a more organized and transparent maintenance management process.

Right now, the infrastructure maintenance process involves multiple parties and several stages, from identifying an issue to completing necessary repairs. Without a centralized system to manage this workflow, several difficulties might arise:

Disorganized issue reporting has always been a problem for an organization whether it's technical or economic issues. Users need a structured way to report the broken equipment, including its description, location and supporting photo. Without a centralized platform for it, organizing information can be difficult.

After receiving the issue, maintenance staff need to wait for approval from the higher ups to assign personnel and oversee the process. Managing these activities without a centralized platform may be too difficult to keep track of all the issues.

Because of that, the primary goal of InfraLine is to provide a centralized web application that simplifies infrastructure issue reporting and improves the management of maintenance activities. The system aims to achieve the following objectives:

Centralized issue reporting is the first and foremost goal, allowing students and workers to issue reports containing title, description, location and a photo of the equipment.

Report moderation is the next phase, allowing administrators to review, approve or reject submitted reports before they can be publicly viewed on the forum.

Maintenance progress tracking is also a crucial part of the process, allowing maintenance personnel to view assigned work orders, carry out maintenance activities and update their statuses with email responses for more convenience.

Work order management is no less important than others as to convert approved reports into work orders accessible to maintenance and management staff, with relevant attributes such as status, location, priority, assignees and equipment type.

InfraLine is designed to operate as a web-based application accessible through a web browser. It supports different categories of users, each with specific responsibilities within the maintenance workflow.

For workers and students, the application is used to submit reports about broken infrastructure, provide information and photos and view approved reports on the forum. For administrators and management staff, they can moderate reports, manage work order attributes, assign maintenance personnel, and oversee maintenance activities. And finally, for maintenance staff, they can access assigned work orders through checklists, carry out maintenance tasks and update the work order statuses.

For the technical side of things, the proposed application will be using the following technologies: For frontend, it's HTML, CSS, React and JavaScript. For the backend, Django (Python) is used for server-side logic and request processing. PostgreSQL is used for storing application data including reports, work orders, user information and maintenance records.

Finally, the design and implementation of InfraLine must comply with the following constraints:

InfraLine must be a web application rather than a desktop application.

The implementation at its core must be Python, HTML, CSS and JavaScript, with PostgreSQL as the database management system.

Different categories of users must have access to functions appropriate to their responsibilities.

Reports must be included with necessary information: title, location, description, other media (optional) and tags (optional).

Submitted reports must be subject to administrator approval before being displayed publicly and are available for work orders.

Each work order must support the necessary attributes, including status, location, priority, and assignees.

These constraints establish technical and functional boundaries that must be considered throughout the design and implementation of InfraLine.

# 4 Stakeholders

| # | Stakeholder | Description | Interests / concerns |
|---|---|---|---|
| 1 | Requesters | Non-management facility users such as employees, students, workers, etc. They are the main beneficiaries of the equipment in the workplace. | - Reporting issues.<br>- Tracking issues. |
| 2 | Technicians | Maintenance personnel responsible for troubleshooting and repairing equipment. | - Tracking work orders.<br>- Carrying out work orders. |
| 3 | Managers | Upper staff who oversee maintenance operations in the facility. | - Approving reports.<br>- Tracking work orders.<br>- Assigning work orders. |
| 4 | IT department | The facility's internal IT team, responsible for configuring and maintaining the system after deployment. | - Setting up user accounts.<br>- Configure settings.<br>- Provide internal support. |
| 5 | Development team | The team responsible for building the software system. | - Coordination with the facility to build the system. |

# 5 Requirements

## 5.1 Functional Requirements

| ID | Requirement | Feature | Priority | Source |
|---|---|---|---|---|
| FR-01 | The system shall allow requesters to upload a post about broken infrastructure on a forum | F01 | Must | Document (PA1) |
| FR-02 | The system shall let requesters ping their location via a dropdown box to let maintenance know where the broken infrastructure is. | F03 | Must | Document (PA1) |
| FR-03 | The system shall allow administrators and maintenance team members to see the current status of a problem | F05 | Must | Document (PA1) |
| FR-04 | The system shall provide maintenance members with a checklist of problems they were assigned. | F06 | Must | Document (PA1) |
| FR-05 | The system shall allow requesters and administrators to set tags to posts to specify the kind of problem infrastructure (mechanical, electric, technological,...). | F08 | Must | Document (PA1) |
| FR-06 | The system shall allow administrators to explicitly set priority to problems posted. | F09 | Must | Document (PA1) |
| FR-07 | The system shall allow administrators to approve or reject a post if they feel the problem is minor or out of the maintenance team's scope, and the requester shall receive an in-app notification whenever their post is approved/rejected. | F10 | Must | Document (PA1) |
| FR-08 | The system shall allow requesters and administrators to see the history of fixed infrastructure, who fixed it and when it was fixed. | F04 | Should | Document (PA1) |
| FR-09 | The system shall send an email response to the requester when the problem has been fixed. | F07 | Could | Document (PA1) |
| FR-10 | The system shall establish a chatbox between a maintenance team member and a requester, letting them chat privately about the problem. | F02 | Could | Document (PA1) |

## 5.2 Non-Functional Requirements

| ID | Category | Requirement (measurable) | How it will be verified |
|---|---|---|---|
| NFR-01 | Performance | Under 10 concurrent users, the forum page shall render all posts in <= 2 seconds at P95. | The team self-testing the app using VMs or Postman. |
| NFR-02 | Usability | Under slow internet connection (2000ms latency or more), any request taking more than 2 seconds shall show a loading indicator within 500 ms of performing an action (transitioning into another page, creating a work order,...). | Throttle network to "3G" in DevTools and test the system's functions (login, creating a new work order, approve/reject a post...). |
| NFR-03 | Security | Under 30 total accounts, 0 of 10 scripted attack attempts (unauthenticated API access, SQL injection, accessing another role's data) shall succeed. | Testing via Postman / Bruno 10 times and recording the results. |
| NFR-04 | Compatibility | On Google Chrome, Edge and Firefox (Windows) all the core features should be working smoothly. | The team self-testing the app and its features on multiple platforms. |
| NFR-05 | Reliability | Under 10 concurrent users on the deployed app, at least 99% of requests shall return non-5xx responses. | k6 or Locust from one laptop; read the status-code distribution in the summary. |
| NFR-06 | Compliance | Under 40 recorded posts, 100% of delete attempts shall archive the post with its maintenance history intact to comply with ISO 55001 asset record-keeping. | Attempt to delete every record via the UI and API, then check the database. |
| NFR-07 | Capacity | Users shall upload attachments of at most 10 MB each, with total attachment storage at most 800 MB. A 10 MB upload shall complete in ≤ 15 s on a 20 Mbps connection. | Upload dummy files; read timings in the DevTools Network tab; verify the quota in the dashboard. |
| NFR-08 | Maintainability | 100% of merges to main shall have at least one approved review from another team member. | GitHub protection rules. |
| NFR-09 | Environmental | Using 5 different common browsers, the app should work smoothly on the latest two versions of Chrome, Firefox, Edge. | Test all of the workflows on each of the browsers. |
| NFR-10 | Scalability | When the Supabase file storage reaches 80% of the free tier, Supabase will be upgraded to the Pro tier within 3 months, upgrading from a 1GB file storage to 100GB file storage. | Confirm the paid plan's storage limit on Supabase's pricing page, and confirm the storage config is environment-variable-based. |

## 5.3 Domain Requirements and Business Rules

| ID | Rule | Source (policy, law, standard) |
|---|---|---|

| | | |
|---|---|---|
| BR-01 | Each user must have exactly one account with permissions based on their role in the workplace. | Policy |
| BR-02 | Users must select a location when posting a report. | Policy |
| BR-03 | Edited reports require re-approval. | Policy |
| BR-04 | Only administrators are allowed to approve reports, set priorities, and assign work orders. | Policy |
| BR-05 | Administrators must provide a reason when rejecting a report. | Policy |
| BR-06 | The status of a work order can only be updated by administrators and the technician(s) assigned to that work order. | Policy |
| BR-07 | Administrators and technicians must provide a reason or evidence of completion when closing a work order. | Policy |
| BR-08 | Infrastructure data must be kept confidential. | Policy |

## 5.4 Acceptance Criteria

- F01: Broken infrastructure report
  + Success

    > Given: Requesters log in to InfraLine and prepare to post a report.

    > When: Requesters write a description of broken infrastructure report, then press submit.

    > Then: Waiting for the approval from administrators. After that, the post will be displayed on the forum if it is approved.
  + Failure

    > Given: Users log in to InfraLine and prepare to post a report.

    > When: The user left the description empty or did not select the location from the dropdown box and clicked the 'Submit' button.

    > Then: The system doesn't allow you to submit the post and shows an error message asking you to enter a description / choose location.

- F03: Location Ping
  + Choose location

    > Given: The user is filling out a form to report broken infrastructure.

    > When: The user selects a specific location from the dropdown list.

    > Then: The system records the selected location and assigns it to the incident report post.
  + Attach an image (optional)

> Given: The user has selected a location from the list and has written a description.

> When: Users attach a relevant image for the report.

> Then: The system attaches that image for the report.

- F05: Maintenance status
  + See maintenance status

    > Given: User or Administrator is viewing the list or details of an incident report.

    > When: The report page has finished loading.

    > Then: The system clearly shows the current status of the issue ("Assigned", "In progress", and "Closed"').

- F06: Maintenance team checklist
  + Show checklist

    > Given: The maintenance team member logs into their account.

    > When: Maintenance team member access to checklist.

    > Then: The system shows the checklist that has been assigned to them to handle.
  + Mark task as complete

    > Given: Maintenance team member is looking at the checklist.

    > When: They check off a task that has been fixed, then they give a reason and attach an image for their decision.

    > Then: The system updates that task to "Closed"' in the checklist.

- F08: Broken infrastructure post tags
  + Post tagging

    > Given: User or administrator is editing/creating an incident report.

    > When: The user/administrator selects the appropriate tags (Mechanical, Electrical, Critical, ...) and saves.

    > Then: The report is attached with selected tags to help the maintenance team evaluate and classify the work.

- F09: Set task priority
  + Priority task

> Given: The admin is looking at a post on the forum.

> When: The admin chooses the priority level for the post (like High, Medium, Low) and confirms.

> Then: The system saves the new priority level and sorts this priority in the maintenance team's checklist.

- F10: Problem moderation
  + Moderate problem

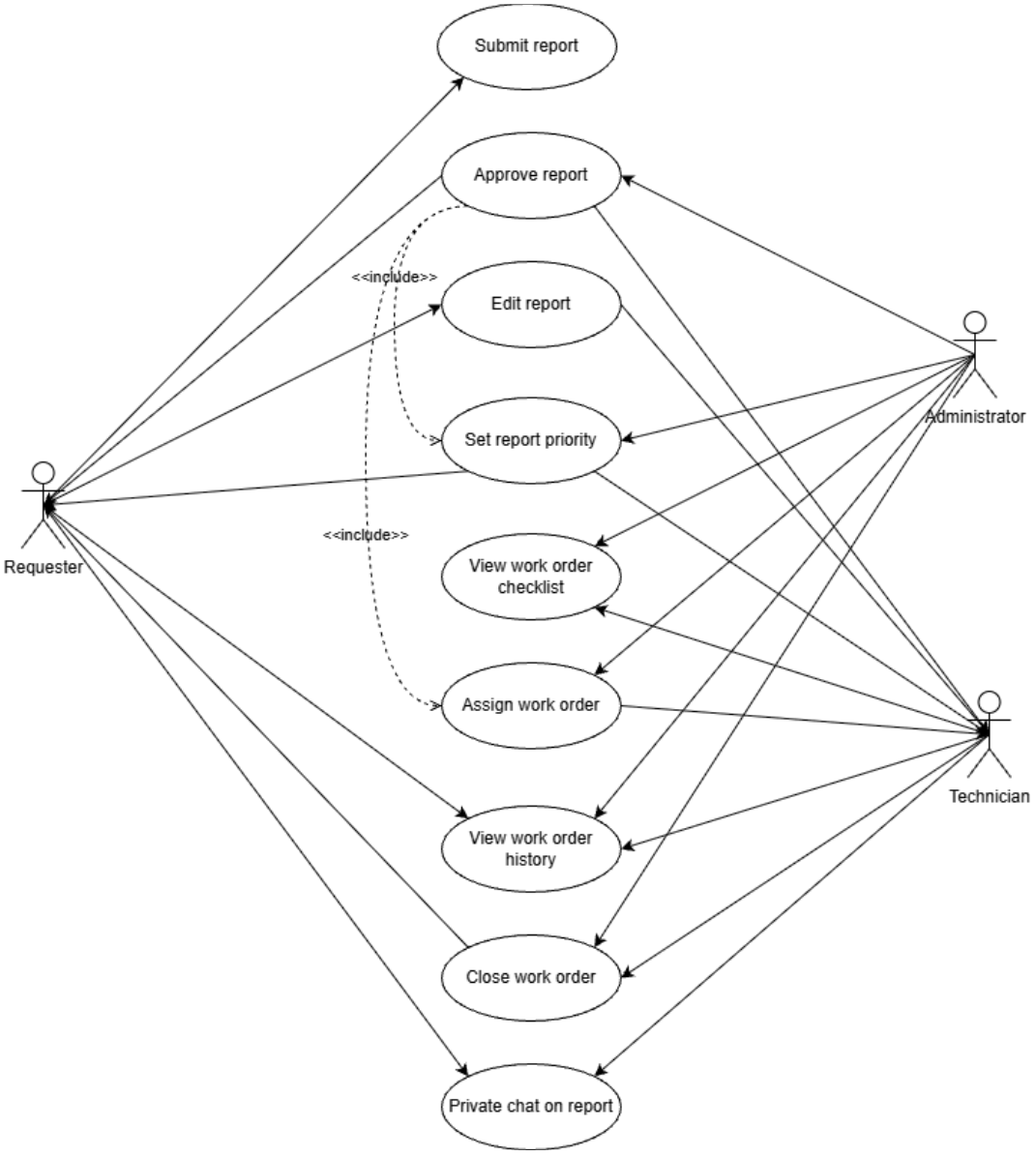
> Given: The newly submitted incident report is pending approval.

> When: The administrator assigns technicians to the report and sets the report’s priority. Then they review and clicks 'Approve' for the approval post.

> Then: The post is publicly displayed on the forum and sent to the maintenance team's checklist.

# 6 Use Case Model

## 6.1 Use Case Diagram




## 6.2 Actors

| Actor | Description |
|---|---|
| Requester | Non-management facility users such as employees, students, workers, etc. |
| Technician | Maintenance personnel responsible for troubleshooting and repairing equipment. |
| Administrator | Includes managers and IT personnel. |

## 6.3 Use Case List

| ID | Use case | Primary actor | Priority | Related FR |
|---|---|---|---|---|
| UC-01 | Requester submits report for review. | Requester | Must | FR-01, FR-02, FR-05 |
| UC-02 | Requester edits uploaded report. | Requester | Must | FR-05 |
| UC-03 | Administrator approves report for upload. | Administrator | Must | FR-07 |
| UC-04 | Administrator assigns a work order to technician. | Administrator | Must | FR-04 |
| UC-05 | Technician and administrator view work order checklists. | Technician, Administrator | Must | FR-03, FR-04 |
| UC-06 | Administrator sets a report's priority. | Administrator | Must | FR-06 |
| UC-07 | Technician or administrator closes a work order. | Technician, Administrator | Must | FR-09 |
| UC-08 | Requester, administrator and technician view the work order history. | Requester, Administrator, Technician | Should | FR-08 |
| UC-09 | Requester engages in a private chat with technician on a report. | Requester, Technician | Could | FR-10 |

# 7 Use Case Specifications

## 7.1 Use Case UC-01

| *Use case ID* | UC-01 |
|---|---|
| *Use case name* | Submit report |
| *Brief description* | The user creates a report in the form of a forum post and submits it for approval. |
| *Primary actor* | Requester |
| *Secondary actors* | |
| *Priority* | Must |
| *Related requirements* | FR-01, FR-02, FR-05 |
| *Trigger* | The user clicks a "Create Report" button. |
| *Pre-conditions* | The user is logged in. |
| *Post-conditions* | A report is submitted for administrator approval. |
| *Main flow* | 1. The user enters the title of the report.<br>2. The user selects the location relevant to the report from a dropdown menu.<br>3. The user enters the description of the report.<br>4. The user attaches relevant media (optional).<br>5. The user selects relevant tags (optional).<br>6. The user clicks "Submit".<br>7. The system verifies if the title and description are within character limits.<br>8. The system verifies if a location has been selected from the dropdown box.<br>9. The system verifies if attachments are within size limits.<br>10. The system creates the report and sends it to a queue for administrators. |
| *Alternative flows* | A1 (at step 7):<br>A1-1. If the title or description is empty or exceeds character limits, the system displays an error message to the user.<br><br>A2 (at step 8):<br>A2-1. If a location has not been selected, the system displays an error message to the user.<br><br>A3 (at step 9):<br>A3-1. If at least one attachment exceeds size limits, the system displays an error message to the user. |
| *Exception flows* | E1:<br>E1-1. If the system encounters an unexpected issue when creating the report, display an error message to the user. |
| *Business rules* | BR-02 |
| *Non-functional constraints* | Steps 7-9 must take less than 1 second.<br>Step 10 must take less than 3 seconds. |

## 7.2 Use Case UC-02

| *Use case ID* | UC-02 |
|---|---|
| *Use case name* | Edit report |
| *Brief description* | The user edits one of their approved reports on the forum. |
| *Primary actor* | Requester |
| *Secondary actors* | Technician |
| *Priority* | Must |
| *Related requirements* | FR-05 |
| *Trigger* | The user clicks an "Edit Report" button. |
| *Pre-conditions* | The user is logged in.<br>The user has a report that has been approved and uploaded to the forum. |
| *Post-conditions* | An edited version of the forum post is submitted for administrator approval. |
| *Main flow* | 1. The user edits whichever field they want except for priority.<br>2. The user clicks "Submit".<br>3. The system verifies if the title and description are within character limits.<br>4. The system verifies if a location has been selected from the dropdown box.<br>5. The system verifies if attachments are within size limits.<br>6. The system creates a new report and sends it to a queue for administrators.<br>7. The system archives the original post from the forums.<br>8. The system archives the associated work order.<br>9. The system sends an in-app notification to the assigned technician(s) that the work order is temporarily halted. |
| *Alternative flows* | A1 (at step 3):<br>A1-1. If the title or description is empty or exceeds character limits, the system displays an error message to the user.<br><br>A2 (at step 4):<br>A2-1. If a location has not been selected, the system displays an error message to the user.<br><br>A3 (at step 5):<br>A3-1. If at least one attachment exceeds size limits, the system displays an error message to the user. |
| *Exception flows* | E1:<br>E1-1. If the system encounters an unexpected issue when creating the report, display an error message to the user. |
| *Business rules* | BR-02, BR-03 |
| *Non-functional constraints* | Steps 3-5 must take less than 1 second.<br>Steps 6-9 must take less than 4 seconds. |

## 7.3 Use Case UC-03

| *Use case ID* | UC-03 |
|---|---|

| | |
|---|---|
| *Use case name* | Approve report |
| *Brief description* | The user approves a report to upload it to the forum. |
| *Primary actor* | Administrator |
| *Secondary actors* | Requester, Technician |
| *Priority* | Must |
| *Related requirements* | FR-07 |
| *Trigger* | The user selects a "Review Reports" button. |
| *Pre-conditions* | The user is logged in as an administrator. |
| *Post-conditions* | A report is uploaded to the forums.<br>A work order is sent to the selected technician(s). |
| *Main flow* | 1. The user is presented with a list of reports awaiting approval.<br>2. The user clicks on a report to see its details.<br>3. The user assigns technicians to the report (UC-04).<br>4. The user sets the report's priority (UC-06).<br>5. The user clicks "Approve".<br>6. The system uploads the report to the forum. |
| *Alternative flows* | A1 (at step 2):<br>A1-1. If the user intends to reject, skip steps 3-4.<br>A1-2. The user enters the reason for rejection and clicks "Reject".<br>A1-3. The system verifies the reason is within character limits.<br>A1-4. The system deletes the report.<br>A1-5. The system sends an in-app notification to the report author with the reason for rejection.<br><br>A1-A (at step A1-3):<br>A1-A-1. If the reason is empty or exceeds the character limit, the system displays an error message to the user.<br><br>A1-B (at step A1-3):<br>A1-B-1. If the report is an edit, the system restores the archived original version to the forum.<br>A1-B-2. The system restores the archived original work order.<br>A1-B-3. The system sends an in-app notification to the assigned technician(s) that the work order is restored.<br><br>A2 (at step 6):<br>A2-1. If the report is an edit, the system appends "[Updated]" to the start of the report title.<br>A2-2. The system deletes the archived original report.<br>A2-3. The system deletes the archived original work order.<br><br>A3 (at step 1):<br>A3-1. If there are no pending reports, the system displays the text "There are no pending reports at this time." |
| *Exception flows* | E1:<br>E1-1. If the system encounters an unexpected issue when uploading, deleting, or restoring the report, display an error message to the user. |
| *Business rules* | BR-03, BR-04, BR-05 |

| | |
|---|---|
| *Non-functional constraints* | Step 6 must take less than 3 seconds.<br>Steps A1-3 to A1-5 must take less than 3 seconds.<br>Steps A1-B-1 to A1-B-3 must take less than 3 seconds. |

## 7.4 Use Case UC-04

| *Use case ID* | UC-04 |
|---|---|
| *Use case name* | Assign work order |
| *Brief description* | The user assigns a report to one or multiple technicians. |
| *Primary actor* | Administrator |
| *Secondary actors* | Technician |
| *Priority* | Must |
| *Related requirements* | FR-04 |
| *Trigger* | The user clicks an "Assign" button on a report. |
| *Pre-conditions* | The user is logged in as an administrator.<br>There exists either a pending or approved report on the forum. |
| *Post-conditions* | A work order is assigned to one or multiple technicians' checklists. |
| *Main flow* | 1. The system presents a pop-up.<br>2. The user selects at least one technician.<br>3. The user enters additional notes (optional).<br>4. The user clicks "Assign".<br>5. The system verifies that at least one technician has been selected.<br>6. The system verifies that the notes are within character limits.<br>7. The system creates a work order with the title, description, attachments, tags, and priority from the report, as well as the notes.<br>8. The system sends the work order to the checklists of the selected technician(s).<br>9. The system updates the report to say which technician(s) the issue has been assigned to. |
| *Alternative flows* | A1 (at step 5):<br>A1-1. If no technicians have been selected, the system displays an error message to the user.<br><br>A2 (at step 6):<br>A2-1. If the notes exceed the character limit, the system displays an error message to the user.<br><br>A3 (at step 8):<br>A3-1. If a technician has been unassigned, the system deletes the work order from their checklist.<br>A3-2. The system sends an in-app notification to the unassigned technician. |
| *Exception flows* | E1:<br>E1-1. If the system encounters an unexpected issue when creating or deleting the work order, display an error message to the user. |
| *Business rules* | BR-04 |

| | |
|---|---|
| *Non-functional constraints* | Steps 5-6 must take less than 1 second.<br>Steps 7-9 must take less than 4 seconds.<br>Steps A3-1 to A3-2 must take less than 3 seconds. |

## 7.5 Use Case UC-05

| *Use case ID* | UC-05 |
|---|---|
| *Use case name* | View work order checklist |
| *Brief description* | The user views the list of work orders assigned to them. |
| *Primary actor* | Technician, Administrator |
| *Secondary actors* | |
| *Priority* | Must |
| *Related requirements* | FR-03, FR-04 |
| *Trigger* | The user selects a "Maintenance Checklist" button. |
| *Pre-conditions* | The user is logged in as a technician or administrator. |
| *Post-conditions* | A checklist of work orders is displayed to the user. |
| *Main flow* | 1. The user is presented with a list of work orders assigned to them.<br>2. The user may click on a work order to view its details and update its status. |
| *Alternative flows* | A1 (at step 1):<br>A1-1. If there are no work orders assigned to the user, the system displays the text "You have no work orders at this time."<br><br>A2 (at step 1):<br>A2-1. If the user is logged in as an administrator, display all work orders in the system. |
| *Exception flows* | E1:<br>E1-1. If the system encounters an unexpected issue when displaying the checklist, display an error message to the user.<br><br>E2:<br>E2-1. If the system encounters an unexpected issue when displaying a work order, display an error message to the user. |
| *Business rules* | BR-06 |
| *Non-functional constraints* | Step 1 and A2-1 must take less than 0.5 seconds per list item. |

## 7.6 Use Case UC-06

| *Use case ID* | UC-06 |
|---|---|
| *Use case name* | Set report priority |
| *Brief description* | The user sets the priority of a report. |
| *Primary actor* | Administrator |
| *Secondary actors* | Requester, Technician |
| *Priority* | Must |
| *Related requirements* | FR-06 |
| *Trigger* | The user clicks a "Set Priority" button. |

| | |
|---|---|
| *Pre-conditions* | The user is logged in as an administrator.<br>There exists either a pending or approved report on the forum. |
| *Post-conditions* | The report has a priority. |
| *Main flow* | 1. The system presents a list of priority levels (e.g., Low, Medium, High).<br>2. The user selects one of the listed priority levels.<br>3. The system assigns the selected priority level to the report. |
| *Alternative flows* | A1 (at step 3):<br>A1-1. If it is an already approved report, the system sends an in-app notification to the author and assigned technician(s) of the change. |
| *Exception flows* | E1:<br>E1-1. If the system encounters an unexpected issue when assigning the priority, display an error message to the user. |
| *Business rules* | BR-04 |
| *Non-functional constraints* | Step A1-1 must take less than 2 seconds. |

## 7.7 Use Case UC-07

| *Use case ID* | UC-07 |
|---|---|
| *Use case name* | Close work order |
| *Brief description* | The user closes a work order. |
| *Primary actor* | Technician, Administrator |
| *Secondary actors* | Requester |
| *Priority* | Must |
| *Related requirements* | FR-09 |
| *Trigger* | The user clicks a "Close Work Order" button. |
| *Pre-conditions* | The user is logged in as a technician or administrator.<br>There exists a work order.<br>• If the user is logged in as a technician, they must be assigned to that work order. |
| *Post-conditions* | The work order is deleted.<br>The associated report is deleted.<br>An email is sent to the report author. |
| *Main flow* | 1. The user enters the reason for closure.<br>2. The user selects "Yes" or "No" on whether the issue has been resolved.<br>3. The user attaches relevant media.<br>4. The user clicks "Close Order".<br>5. The system verifies if a reason has been specified.<br>6. The system verifies if attachments are within size limits.<br>7. The system deletes the work order from all checklists.<br>8. The system deletes the associated report from the forum.<br>9. (Could have) The system sends an email to the report author with the contents of the original report, the assigned technicians (if any), the reason for closure, and attachments. |
| *Alternative flows* | A1 (at step 2):<br>A1-1. If the user selects "No", step 3 is optional.<br><br>A2 (at step 5): |

| | |
|---|---|
| | A2-1. If a reason has not been specified, the system displays an error message to the user.<br><br>A3 (at step 6):<br>A3-1. If at least one attachment exceeds size limits, the system displays an error message to the user. |
| *Exception flows* | E1:<br>E1-1. If the system encounters an unexpected issue when displaying the checklist, display an error message to the user.<br><br>E2:<br>E2-1. If the system encounters an unexpected issue when deleting the work order and report, display an error message to the user.<br>E2-2. Skip step 9.<br><br>E3:<br>E3-1. If the system encounters an unexpected issue when sending the email, display an error message to the user. |
| *Business rules* | BR-06, BR-07 |
| *Non-functional constraints* | Steps 5-6 must take less than 1 second.<br>Steps 7–9 must take less than 3 seconds. |

# 8 Prototype / Mockups

| Screen | Related use case(s) | Mockup / link |
|---|---|---|
| S01 – Forum | UC-01 | Requester perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-3&t=Y1ZpNMRa7AIYOb1H-4<br>Administrator perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-958&t=Y1ZpNMRa7AIYOb1H-4<br>Technician perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-2246&t=Y1ZpNMRa7AIYOb1H-4 |
| S02 – Create report | UC-01 | https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-289&t=Y1ZpNMRa7AIYOb1H-4 |
| S03 – Report detail | UC-02, UC-04, UC-06, UC-08, UC-09 | Requester perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=35-453&t=ulI7BkNRKBeeZYxQ-4<br>Administrator perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-1163&t=Y1ZpNMRa7AIYOb1H-4<br>Technician perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=5-3020&t=Y1ZpNMRa7AIYOb1H-4 |
| S04 – Edit report | UC-02 | https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-594&t=Y1ZpNMRa7AIYOb1H-4 |
| S05 – Work order history | UC-08 | Requester perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-746&t=Y1ZpNMRa7AIYOb1H-4<br>Administrator perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-1307&t=Y1ZpNMRa7AIYOb1H-4<br>Technician perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-2458&t=Y1ZpNMRa7AIYOb1H-4 |
| S06 – Report moderation | UC-03, UC-04, UC-06 | https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-1487&t=Y1ZpNMRa7AIYOb1H-4 |
| S07 – Assign work order (pop-up) | UC-04 | https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-1487&t=Y1ZpNMRa7AIYOb1H-4 |

| | | |
|---|---|---|
| | | (opens by clicking "+ Assign technician(s)" in S06)<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-1163&t=Y1ZpNMRa7AIYOb1H-4<br>(opens by clicking "Reassign technician(s)" in S03) |
| S08 – Maintenance checklist | UC-05 | Technician perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-2644&t=Y1ZpNMRa7AIYOb1H-4<br>Administrator perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-1716&t=Y1ZpNMRa7AIYOb1H-4 |
| S09 – Work order detail | UC-05, UC-07, UC-09 | Technician perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=35-2&t=ulI7BkNRKBeeZYxQ-4<br>Administrator perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-2025&t=Y1ZpNMRa7AIYOb1H-4 |
| S10 – Close work order (pop-up) | UC-07 | Technician perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=35-2&t=ulI7BkNRKBeeZYxQ-4<br>Administrator perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=1-2025&t=Y1ZpNMRa7AIYOb1H-4<br>(opens by clicking "Close Work Order" in S09) |
| S11 – Messages (private chat) | UC-09 | Requester perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=35-293&t=ulI7BkNRKBeeZYxQ-4<br>Technician perspective:<br>https://www.figma.com/proto/VTWOeq6dMZwBV5kLWom2PA/Untitled?node-id=35-120&t=ulI7BkNRKBeeZYxQ-4 |

Basic accessibility check: contrast is clear and readable, with dark gray text on white backgrounds. As for labels, input fields and buttons are labeled. Keyboard use could not be tested because the Figma prototype doesn't support keyboard navigation.

# 9 Requirements Traceability Matrix

| Req. ID | Use case | Screen | Design element (PA3) | Test cases (PA4) |
|---|---|---|---|---|
| FR-01 | UC-01 | S01, S02 | | |
| FR-02 | UC-01 | S01, S02 | | |
| FR-03 | UC-05 | S08, S09 | | |
| FR-04 | UC-04, UC-05 | S03, S06, S07, S08, S09 | | |
| FR-05 | UC-01, UC-02 | S01, S02, S03, S04 | | |
| FR-06 | UC-06 | S03, S06 | | |
| FR-07 | UC-03 | S06 | | |
| FR-08 | UC-08 | S03, S05 | | |
| FR-09 | UC-07 | S09, S10 | | |
| FR-10 | UC-09 | S03, S09, S11 | | |

# 10 Requirements Review

Who took part: Hoàng Minh Huy, Nguyễn Chí Minh, Đoàn Minh Tuấn, Lê Nguyễn Quốc Bảo, Nguyễn Thế Huy.

Date: 2/11/2026.

| # | Issue found | Requirement(s) | Resolution |
|---|---|---|---|
| 1 | The mistake flow: Approve -> assign technician -> set priority. | FR-07 | The current flow: Assign technician -> set priority -> approve |
| 2 | We did not decide to send notification about the reason for rejection/approval to the requester. | FR-07 | The requester shall receive an in-app notification whenever their post is approved/rejected. |
| 3 | At first, we did not mention the security. | BR-08 | Now, the infrastructure data must be confidential. |
| 4 | Initially, the requirement is "The app shall offer Light, Dark, and System-default themes. Switching shall apply to 100% of pages without reload and persist across sessions. " but we think it refers to the UI design, so we decided to change it. | NFR-02 | The current requirement is "Under slow internet connection (2000ms latency or more), any request taking more than 2 seconds shall show a loading indicator within 500 ms of performing an action (transitioning into another page, creating a work order,...)." |

# 11 AI Usage Log

| # | Member | Tool | Task / purpose | How the output was verified and changed |
|---|---|---|---|---|
| 1 | 24127329 | Claude Sonnet 5.5 | Research the available non-functional requirements for each category. | Reviewed with the BA and Team Leader, as well as the same AI Chatbot to eliminate immeasurable metrics or unrelated non-functional requirements. |
| 2 | 24127450 | Claude Opus 5.5 | Generating the wireframes for Section 8 as HTML files based on our use case list, then importing them into Figma with the html.to.design plugin in Figma. | Checked every screen against the use cases (UC-01 – UC-09), discussed and changed several screens, fixed the imported frames and made the prototype links in Figma. |
| 3 | 24127048 | ChatGPT-5.6 Luna | Research domain requirements and business rules. | Checked sources cited to see if they line up with what the LLM said. Used available documentation to check if the suggestions were suitable for the project. |